

EX 4.5 - 5

a. The constraints of an LP can be written as

$$Ax \leq b \quad A \in \mathbb{R}^{n \times m} \quad b \in \mathbb{R}^m$$

let $s_1, \dots, s_n$ are feasible sol

$s = \sum_{i=1}^n w_i s_i$, $\sum_{i=1}^n w_i = 1$, $w_i \geq 0$ $\forall i$ is convex combination of the feasible sol

$$\begin{aligned} \Rightarrow A s &= A \sum_{i=1}^n w_i s_i = \sum_i w_i A s_i \leq \sum_i w_i b \leq b && \text{FUNCTIONAL CONSTRAINTS } \checkmark \\ \sum_i w_i s_i &\geq 0 && \text{NON-NEGATIVITY CONSTRAINTS } \checkmark \end{aligned}$$

b. some steps for basic feasible sol

EX 3.5.3

$$\max Z = 300A + 250B + 200C$$

$$2A + 3B + 5C \leq 4000$$

$$5A + 2B + 4C \leq 4000$$

$$A, B, C \geq 0$$

AUGMENTED FORM: $\max Z$

$$Z - 300A - 250B - 200C$$

$$2A + 3B + 5C + D = 4000$$

$$5A + 2B + 4C + E = 4000$$

$$A, B, C, D, E \geq 0$$

ITER	BV	Z	A	B	C	D	E	RHS
0	Z	1	-300	-250	-200	0	0	0
	D	0	2	3	5	1	0	4000
	E	0	5	2	4	0	1	4000
1	Z	1	0	-130	40	0	60	240000
	D	0	0	$\frac{11}{5}$	$\frac{19}{5}$	1	$-\frac{2}{5}$	2400
	A	0	1	$\frac{2}{5}$	$\frac{4}{5}$	0	$\frac{1}{5}$	800
2	Z	1	0	0	$\frac{265}{11}$	$\frac{65}{11}$	$\frac{92}{11}$	$\frac{420000}{11}$
	B	0	0	1	$\frac{12}{11}$	$\frac{3}{11}$	$-\frac{2}{11}$	$\frac{12000}{11}$
	A	0	1	0	$\frac{16}{11}$	$\frac{2}{11}$	$\frac{15}{11}$	$\frac{20000}{11}$

$$A = \frac{20000}{11} \approx 363$$

$$B = \frac{12000}{11} \approx 1090$$

$$Z = \frac{420000}{11} \approx 381818$$

4.4-5

$$\max z = 5x_1 + 9x_2 + 7x_3$$

$$x_1 + 3x_2 + 2x_3 \leq 10$$

$$3x_1 + 4x_2 + 2x_3 \leq 12$$

$$2x_1 + x_2 + 2x_3 \leq 8$$

$$x_1, x_2, x_3 \geq 0$$

$$\text{AUGMENTED FORM: } \max z = 5x_1 + 9x_2 + 7x_3 \quad (0)$$

$$x_1 + 3x_2 + 2x_3 + x_4 = 10 \quad (1)$$

$$3x_1 + 4x_2 + 2x_3 + x_5 = 12 \quad (2)$$

$$2x_1 + x_2 + 2x_3 + x_6 = 8 \quad (3)$$

$$x_j \geq 0 \quad j = 1 - 6$$

ITER	BASIC VAR	EQ	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
0	Z	(0)	1	-5	-9	-7	0	0	0	0
	x_4	(1)	0	1	3	2	1	0	0	10
	x_5	(2)	0	3	4	2	0	1	0	12
	x_6	(3)	0	2	1	2	0	0	1	8
1	Z	(0)	1	7/4	0	-5/2	0	9/4	0	27
	x_4	(1)	0	-5/4	0	1/2	1	-3/4	0	1
	x_2	(2)	0	3/4	1	1/2	0	1/4	0	3
	x_6	(3)	0	5/4	0	3/2	0	-1/4	1	5
2	Z	(0)	1	-9/2	0	0	5	-3/2	0	32
	x_3	(1)	0	-5/2	0	1	2	-3/2	0	2
	x_2	(2)	0	2	1	0	-1	1	0	2
	x_6	(3)	0	5	0	0	-3	2	1	2

ITER	BASIC VAR	EQ	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
3	Z	(0)	1	0	0	0	$23/10$	$3/10$	$9/10$	$169/5 = 33,8$
	x_3	(1)	0	0	0	1	$1/2$	$-1/2$	$1/2$	3
	x_2	(2)	0	0	1	0	$1/5$	$1/5$	$-2/5$	$6/5$
	x_1	(3)	0	1	0	0	$-3/5$	$2/5$	$1/5$	$2/5$

OPTIMAL! NO NEGATIVE COEFF IN ROW (0)!

$$x_1 = 2/5$$

$$x_2 = 6/5$$

OPTIMAL SOL!

$$x_3 = 3$$

LAST YEAR EXAM (2021)

$$\max z = 2x_1 + 3x_2$$

$$x_1 + 2x_2 \leq 20$$

$$x_1 + x_2 \leq 12$$

$$4x_1 + 3x_2 \leq 44$$

$$x_1, x_2 \geq 0$$

AUGMENTED FORM: $\max z$

$$z - 2x_1 - 3x_2 = 0$$

$$x_1 + 2x_2 + x_3 = 20$$

$$x_1 + x_2 + x_4 = 12$$

$$4x_1 + 3x_2 + x_5 = 44$$

$$x_j \geq 0 \quad j=1-5$$

ITER	BV	Z	x_1	x_2	x_3	x_4	x_5	RHS
0	Z	1	-2	-3	0	0	0	0
	x_3	0	1	2	1	0	0	20
	x_4	0	1	1	0	1	0	12
	x_5	0	4	3	0	0	1	44
1	Z	1	-1/2	0	3/2	0	0	30
	x_2	0	1/2	1	1/2	0	0	10
	x_4	0	1/2	0	-1/2	1	0	2
	x_5	0	5/2	0	-3/2	0	1	14
2	Z	1	0	0	1	1	0	32
	x_2	0	0	1	1	-1	0	8
	x_1	0	1	0	-1	2	0	4
	x_5	0	0	0	1	-5	1	4

$$x_1 = 4 \quad x_2 = 8 \quad z = 32$$

EX 4.4.8

$$\max Z = -x_1 + x_2 + 2x_3$$

$$\text{st } x_1 + 2x_2 - x_3 \leq 20$$

$$-2x_1 + 4x_2 + 2x_3 \leq 60$$

$$2x_1 + 3x_2 + x_3 \leq 50$$

$$x_i \geq 0 \quad i = 1, 2, 3$$

AUGMENTED FORM: $\max Z = -x_1 + x_2 + 2x_3$

$$x_1 + 2x_2 - x_3 + x_4 = 20$$

$$-2x_1 + 4x_2 + 2x_3 + x_5 = 60$$

$$2x_1 + 3x_2 + x_3 + x_6 = 50$$

$$x_j \geq 0 \quad j = 1 - 6$$

ITER	BV	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
0	Z	1	1	-1	-2	0	0	0	0
	x_4	0	1	2	-1	1	0	0	20 IGNORE
	x_5	0	-2	4	2	0	1	0	60 $60/2 = 30$
	x_6	0	2	3	1	0	0	1	50 $50/1 = 50$
1	Z	1	-1	3	0	0	1	0	60
	x_4	0	0	4	0	1	1/2	0	50
	x_3	0	-1	2	1	0	1/2	0	30
	x_6	0	3	1	0	0	-1/2	1	20 ONLY OPTION
2	Z	1	0	$\frac{10}{3}$	0	0	$\frac{5}{6}$	$\frac{1}{3}$	$\frac{200}{3}$
	x_4	0	0	4	0	1	1/2	0	50
	x_3	0	0	2/3	1	0	1/3	1/3	110/3
	x_1	0	1	1/3	0	0	-1/6	1/3	20/3

$$x_1 = \frac{20}{3} \quad x_2 = 0 \quad x_3 = \frac{110}{3} \quad Z = \frac{200}{3}$$